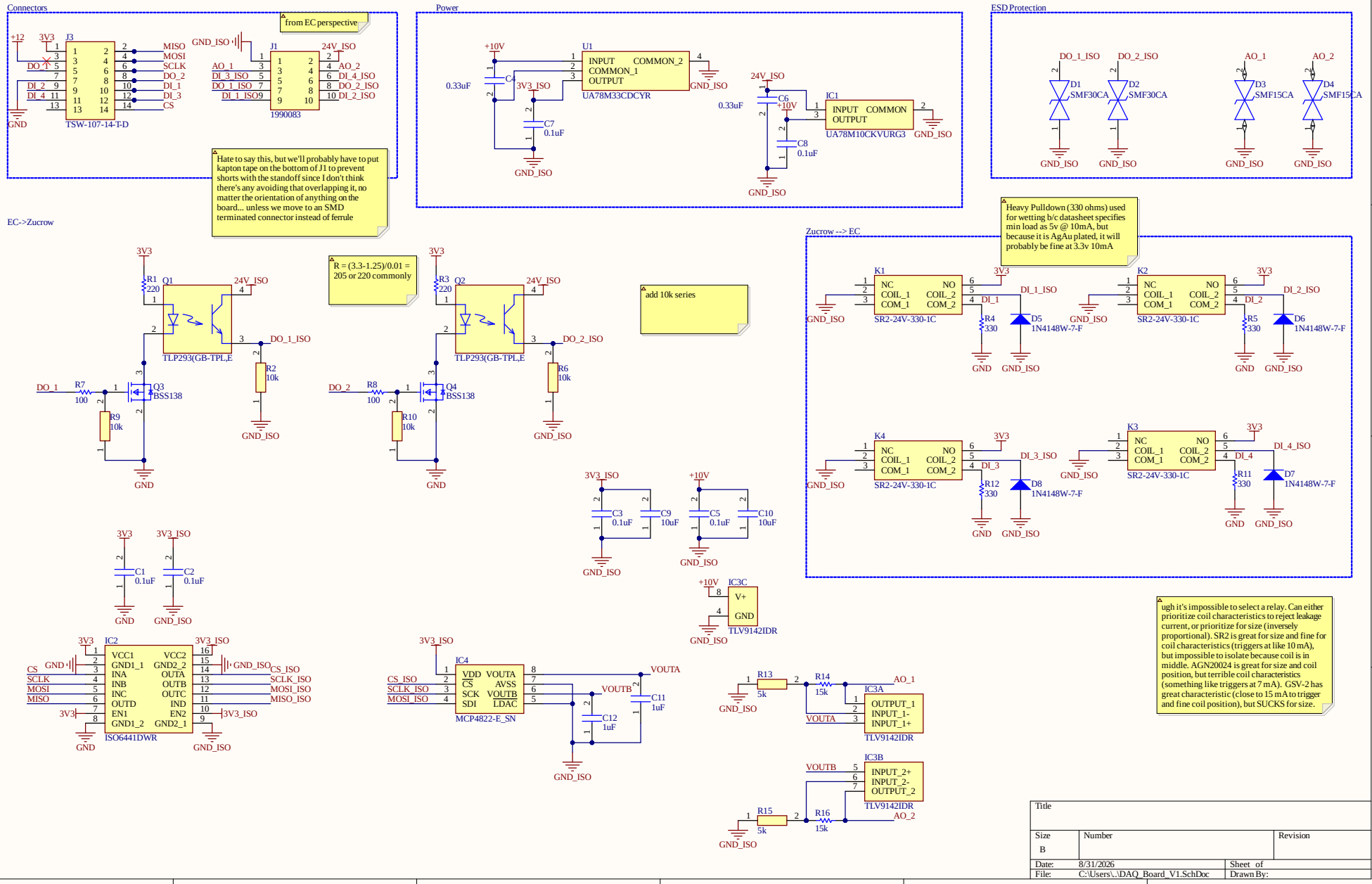
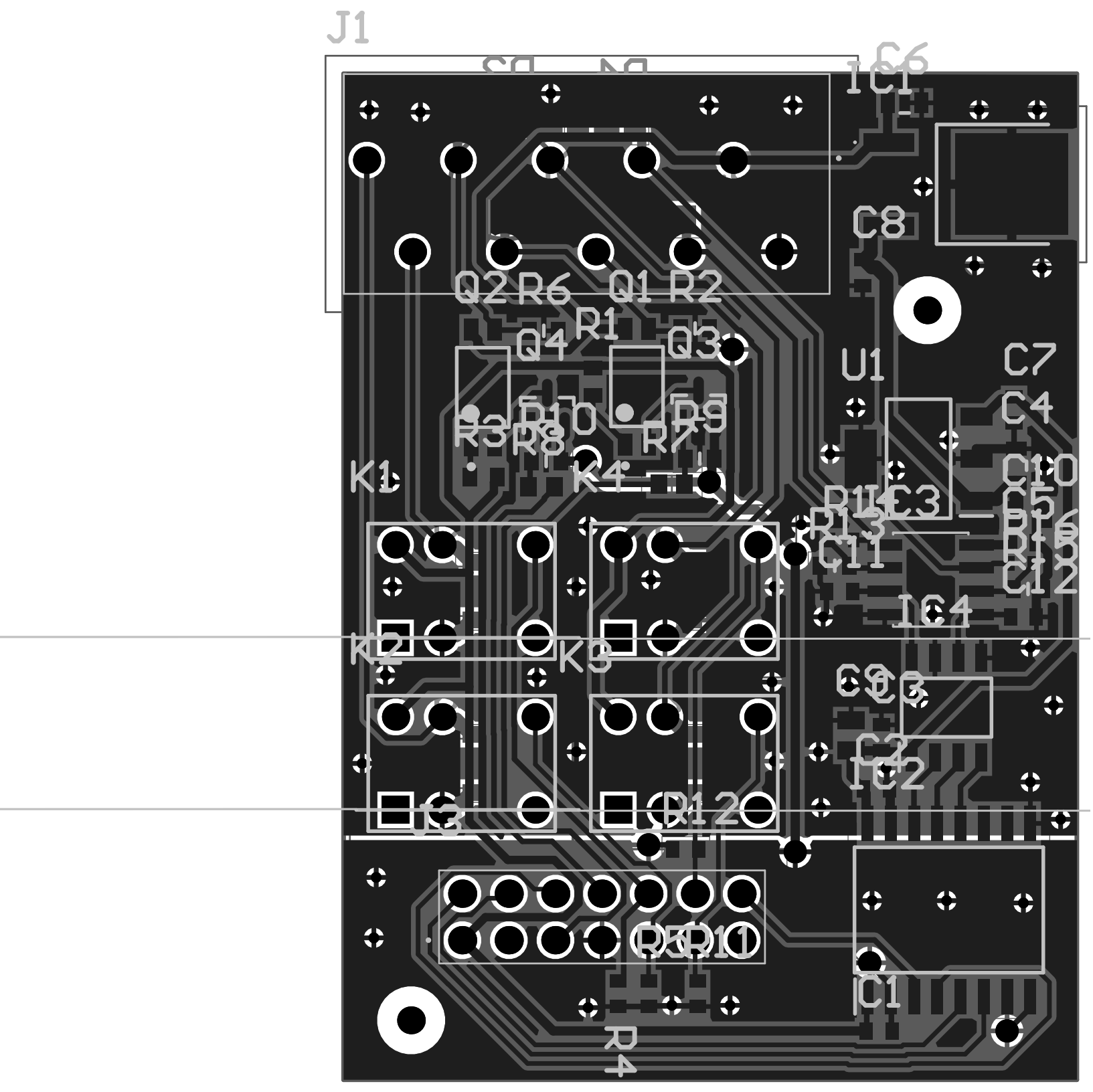

Connectors
from EC perspective
J3
TSW-107-14-T-D
J1
1990083
Hate to say this, but we'll probably have to put kapton tape on the bottom of J1 to prevent shorts with the standoff since I don't think there's any avoiding that overlapping it, no matter the orientation of anything on the board... unless we move to an SMD terminated connector instead of ferule
Power
U1
UA78M33CDCYR
IC1
UA78M10CKVURG3
ESD Protection
D1 SMF30CA
D2 SMF30CA
D3 SMF15CA
D4 SMF15CA
EC->Zucrow
R = (3.3-1.25)/0.01 = 205 or 220 commonly
add 10k series
Q1 TLP293(GB-TPL,E
Q2 TLP293(GB-TPL,E
Q3 BSS138
Q4 BSS138
Zucrow --> EC
Heavy Pulldown (330 ohms) used for wetting b/c datasheet specifies min load as 5v @ 10mA, but because it is AgAu plated, it will probably be fine at 3.3v 10mA
K1
K2
K3
K4
SR2-24V-330-1C
D5 1N4148W-7-F
D6 1N4148W-7-F
D7 1N4148W-7-F
D8 1N4148W-7-F
IC2
ISO6441DWR
IC4
MCP4822-E_SN
IC3A
IC3B
IC3C
TLV9142IDR
ugh it's impossible to select a relay. Can either prioritize coil characteristics to reject leakage current, or prioritize for size (inversely proportional). SR2 is great for size and fine for coil characteristics (triggers at like 10 mA), but impossible to isolate because coil is in middle. AGN20024 is great for size and coil position, but terrible coil characteristics (something like triggers at 7 mA). GSV-2 has great characteristic (close to 15 mA to trigger and fine coil position), but SUCKS for size.
Title
Size B
Number
Revision
Date: 8/31/2026
Sheet of
File: C:\Users\..\DAQ_Board_V1.SchDoc
Drawn By:


J1
IC1
C6
C8
Q2
R6
Q1
R2
R1
Q4
Q3
U1
C7
C4
R3
R8
R10
R7
R9
K1
K4
C10
C5
R14
C3
R13
R16
C11
C12
IC4
K2
K3
C9
C2
IC2
R12
R5
R11
C1
R4

Name	Description	Designator	Footprint	LibRef	Quantity
0.1uF	Capacitor	C1, C2, C3, C5, C7, C8	CAPC1608X90N	CL10B104KB8NNNC	6
0.33uF	Cap Ceramic 330nF 50V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C4, C6	FP-CGA3E-0_3-0_1-IPC_B	CMP-08246-003263-1	2
10uF	Capacitor	C9, C10	CAPC2012X145N	GRM21BC8YA106KE11L	2
1uF	Capacitor	C11, C12	CAPC1608X90N	CL10B105KA8NNNC	2
SMF30CA	TVS Diode (Bi-directional)	D1, D2	SMF30CA	SMF30CA	2
SMF15CA	TVS Diode (Bi-directional)	D3, D4	SODFL3618X108N	SMF15CA	2
1N4148W-7-F	Fast Switching Diode, 100 V, 0.15 A, -65 to 150 degC, 2-Pin?SOD123, RoHS, Tape and Reel	D5, D6, D7, D8	DIOD-SOD123-2_V	CMP-2000-04932-2	4
UA78M10CKVURG3	Integrated Circuit	IC1	UA78M10CKVURG3	UA78M10CKVURG3	1
ISO6441DWR	Integrated Circuit	IC2	SOIC127P1030X255-16N	ISO6441DWR	1
TLV9142IDR		IC3	TLV9142IDR	TLV9142IDR	1
MCP4822-E_SN	Integrated Circuit	IC4	SOIC127P600X175-8N	MCP4822-E_SN	1
1990083	Connector	J1	1990083	1990083	1
TSW-107-14-T-D	Connector	J3	TSW-107-XX-YY-D	TSW-107-14-T-D	1
SR2-24V-330-1C	Relay or Contactor	K1, K2, K3, K4	SR224V3301C	SR2-24V-330-1C	4
TLP293(GB-TPL,E	OPTOISOLATOR 3.75KV TRANS 4-SO	Q1, Q2	FP-11-3C1-MFG	CMP-35887-000010-1	2
BSS138	MOSFET N-CH 50V 220MA SOT-23	Q3, Q4	FP-318-08-IPC_B	CMP-07173-000101-1	2
220	General Purpose Chip Resistors, 220 Ohm, +/- 1%, -55 to 155 degC, 0603 (1608 Metric), RoHS, Tape and Reel	R1, R3	RESC1608X55X25ML10T15	CMP-1659-00018-2	2
10k	Resistor	R2, R6, R9, R10	RESC1608X55N	RC0603FR0710KL	4
330	Chip Resistor, 330 Ohm, +/- 1%, 0.1 W, -55 to 155 degC, 0603 (1608 Metric), RoHS, Tape and Reel	R4, R5, R11, R12	RESC1608X55X25NL10T15	CMP-1659-00020-2	4
100	Chip Resistor, 100 Ohm, +/- 1%, 100 mW, -55 to 155 degC, 0603 (1608 Metric), RoHS, Tape and Ree	R7, R8	RESC1609X50X30ML8T20	CMP-2000-03824-5	2
5k	Resistor	R13, R15	RESC1608X55N	RT0603BRE075KL	2
15k	Chip Resistor, 15 KOhm, +/- 1%, 0.1 W, -55 to 155 degC, 0603 (1608 Metric), RoHS, Tape and Reel	R14, R16	RESC1608X55X25LL10T15	CMP-1659-00032-1	2
UA78M33CDCYR	Undefined or Miscellaneous	U1	SOT230P700X180-4N	UA78M33CDCYR	1